Using the GS130W Stereo Camera on RDK S600

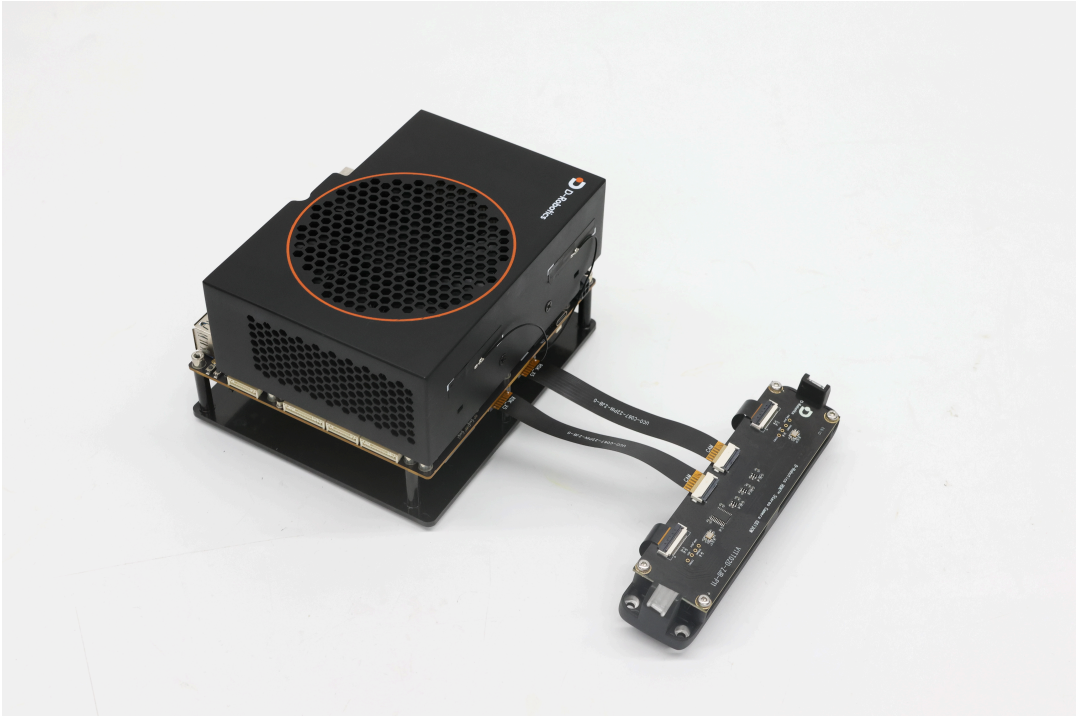
The GS130W stereo camera module is used on RDK S600 through the RDK Stereo Camera SDK. The SDK command-line tool `dsc` is installed together with the core layer and is used to capture stereo images and verify the data path.

1. Environment Preparation

- The sensor supported by this tutorial: **GS130W**
- Install the SDK core layer

  ```
  git clone https://github.com/D-Robotics/RDK-Stereo-Camera-SDK.git
  cd RDK-Stereo-Camera-SDK/core
  make RDKS600
  sudo dpkg -i out/dstereo-cam_*+RDKS600_*.deb
  ```

- Connect the GS130W module to the two MIPI camera interfaces of RDK S600 with two 22-pin cables. **Operate with the power off**:
  - Module side: **the gold fingers face away from the module PCB** (the contacts point away from the PCB); insert horizontally and press down the locking latch
  - Board side: **the gold fingers face the board PCB**; insert horizontally and press down the locking latch



2. Running Method

Command-line tool

```
dsc help                                    # List available commands
dsc version                                 # SDK version and build platform
dsc detect eeprom -b 4 5                    # Scan the I2C buses and addresses
dsc shell -d GS130W                         # Enter interactive debugging
```

After entering interactive mode with `dsc shell`, `run` captures stereo images in real time and prints the status at 50 Hz, `Ctrl-C` stops it, and `exit` quits.

C

```c
#include "dstereo_cam.h"
#include "dstereo_cam_define.h"

dsc_config_t config = DSC_CONFIG(
    "GS130W", DSC_CAMERA_MODE_RECT, 1280, 1088, 30, 200);

dsc_device_t *device = dsc_create();
if (dsc_init(device, &config) != DSC_OK || dsc_start(device) != DSC_OK) {
    dsc_stop(device);
    dsc_deinit(device);
    dsc_destroy(device);
    return 1;
}
```

```
sunrise@ubuntu:~$ gcc demo.c -ldstereocam -o demo
```

Python

```python
import dstereo_cam

config = dstereo_cam.preset(
    "GS130W", dstereo_cam.CameraMode.RECT, 1280, 1088, 30, 200
)
with dstereo_cam.Device(config) as device:
    device.start()
    while device.available_camera() == 0:
        pass
    left = device.read_image()["left"]
```

```
sunrise@ubuntu:~$ python3 demo.py
```

ROS 2

```
source /opt/ros/jazzy/setup.bash
ros2 launch dstereo_cam dstereo_cam.launch.py             # Images, calibration,
and TF
ros2 launch dstereo_cam dstereo_cam_websocket.launch.py   # Same as above,
previewable in a browser
ros2 launch dstereo_cam dstereo_cam_stereonet.launch.py   # View stereo depth on
the same page
```

Precautions

Camera plugging and unplugging precautions

It is strictly forbidden to plug or unplug the camera while the board is powered on; otherwise the camera module can be easily burned out.

On the GS130W module side, the cable is inserted with **the contacts facing away from the module PCB**, which is **the opposite direction** of other common MIPI modules (contacts facing the PCB). Be sure to confirm this when wiring; if connected in the wrong direction, the device cannot be recognized.

- The calibration data is stored in the EEPROM on the module board and is read automatically by the SDK, so no manual calibration is required; in `rect` mode, distortion removal and epipolar rectification are done by the GDC hardware.
- In `raw` mode the output width and height must be 1280 × 1088; common resolutions in `rect` / `resize` mode are 1280×1088, 640×544, and 320×272.